

ECO475 Term Paper: The Effects of Labour Participation on Real GDP Per Capita

Nipun Jaiswal and Ana Elisa Lopez-Miranda*
Department of Economics, University of Toronto
ECO475: Applied Econometrics II
Professor Eduardo Augusto Souza Rodrigues

April 24, 2026

Abstract

Canada's real gross domestic product (GDP) per capita is much lower than that of countries of similar size. This prompts the question if population, or labour participation rate, might explain this. In this study, we use datasets from 1995 - 2024 from Statistics Canada and the Federal Reserve of Economic Data to look at GDP per capita, labour force participation rate, investment in inventories, and labour productivity to find the relationship between GDP per capita and labour force participation rate. We find that labour force participation rate has a significant association with real GDP per capita with our model suggesting that one percentage point increase in participation is associated with \$1,806.88 increase in quarterly GDP per capita growth.

*Contact email: nipun.jaiswal@mail.utoronto.ca a.lopez.miranda@mail.utoronto.ca. I would like to thank Professor Eduardo for his guidance and support throughout this project. All remaining errors are our own.

1 Introduction

Gross Domestic Product (GDP) is a standard measure of economic activity and is often used as a proxy for well-being. Many factors are known to affect it, including consumption, investment, government expenditure, and net exports. When compared to countries of similar size, Canada's GDP is found to be much lower. This prompts the question as to why. Given Canada's geographical size versus population size, the question is if population, or labour participation might be the reason why. In this study, we use datasets from 1995 - 2024 from Statistics Canada and the Federal Reserve of Economic Data to look at GDP per capita, labour force participation rate, investment in inventories, and labour productivity to find the relationship between GDP per capita and labour force participation rate. We find that labour force participation rate has a significant association with real GDP per capita with our model suggesting that one percentage point increase in participation is associated with \$1,806.88 increase in quarterly GDP per capita growth.

The discussion surrounding real GDP per capita and labour force participation rate is particularly precarious as this is a case of simultaneity. This is because, while higher participation likely drives growth, a strong economy (reflected in rising GDP) also creates the incentives and wage growth that pulls more Canadians into the labour force. In other words, each may influence the other. As real GDP per capita increases, so does the labour force participation rate and as the labour force participation rate increases, so does real GDP per capita. This results in endogeneity and biased ordinary least squares estimates. Therefore, the results and conclusions will be in terms of correlation and not impact.

The Government of Canada states on their site (Statistics Canada (2024a)) that the National Gross Domestic Product (GDP) by Income and by Expenditure Accounts measures the production of goods and services in the economy, the incomes from the production and the expenditure on the production. Analysts then use it to form interpretations of productive activity, income, expenditure, saving, investment, financing and wealth. In other words, GDP is the value of goods and services produced that is available for final domestic consumption, investment or export, and is representative of the volume of economic activity in a given period.

Although multifactor productivity is known to improve GDP (Wölfl and Hajkova (2007)), there is a high level of complexity to measure multifactor productivity, and it requires more data re-

quirements. Labour productivity, on the other hand, is simpler to gather and is of interest since it provides more current information than the annual productivity data. It is also necessary to analyze the behaviour of the economy during business cycles and it provides current information Canada's productivity compared to the U.S.

Real GDP per capita is important as this is used in the calculation of demographic, social, and economic indicators. Estimated population helps determine the amount of federal-provincial-territorial transfers. The data also helps determine the annual level of immigration by the government of Canada.

The paper will proceed as follows: section 2 will discuss the literature review and economic intuition as to the variables chosen; section 3 will discuss the institutional background detailing where the data is from and its collection; section 4 will discuss the data with description and summary statistics; section 5 discusses the empirical framework and the empirical strategy; section 6 does the visual checks, unit root test, and regression. section 7 concludes; section A contains the figures; And section B contains the tables.

2 Literature Review

Smith (2024), in *Accounting for the Decline of Canada's real GDP per capita since Mid-2022*, mentions how there are six factors influencing real GDP per capita: labour productivity, average hours worked per job, the employment rate of the working-age population, the ratio of the working-age population to the permanent resident population, the share of permanent residents in the total population, and the relative rate of growth of the government and non-profit sectors compared to that of the business sector. He notes that our GDP fell behind the U.S. in 1990-1992 with another drop in 2015 due to falling world energy prices. The gap continued to widen as Canada's productivity growth slowed and the U.S. productivity grew more quickly. He says that the most important driver of GDP is business sector labour productivity. Another factor that can have influence is average hours worked per business sector job. As people work more or less hours, this will directly affect GDP per capita. Another factor is the number of business sector jobs as a ratio to the working-age population. which is the number of jobs relative to the available people of working age. Another factor is the ratio of working-age population to the permanent-resident

population. All of these contribute to labour force productivity being a contributor to real GDP per capita. Maynard (2007), in *A Comparison of GDP per Capita in Canada and the United States from 1994 to 2005*, decomposed the gap between GDP per Capita in Canada and the U.S. into two parts: labour productivity and labour market conditions, showing that labour productivity separately affects GDP as compared to labour participation rate. It also lends to the idea that perhaps labour participation rate has a correlation with GDP per capita.

Cao and Kozicki (2017), in *Real GDI, Productivity, and the Terms of Trade in Canada*, described how total factor productivity and the terms of trade affect the growth of real gross domestic income. Again, supporting the idea that labour productivity affects real GDP per capita.

Baldwin et al. (2004), in *Catching Up and Falling Behind: The Performance of the Provincial GDP per Capita from 1990 to 2003*, looked at whether increases in real GDP per capita was related to changes in productivity or changes in labour market conditions. They decomposed GDP per capita into components relating to labour productivity, hours worked per person, the percentage of eligible people who are employed, and the percentage of the population who are of working age. They found that provinces with high growth rates in real GDP per capita also had high growth in labour productivity. Again showing that labour productivity separately affects GDP as compared to labour participation rate.

Hornstein (1998) in *Inventory Investment and the Business Cycle* report that the business cycle is very dependent on the changes in inventory investment. He writes “changes in inventory investment apparently account for a substantial fraction of changes in gross domestic product (GDP)“. Blinder and Maccini (1991) in *Taking stock: a critical assessment of recent research on inventories* say that in a U.S. recession, declining inventory investments account for the majority of the decline in GDP. Thus, supporting the idea that investment in inventories affect real GDP per capita.

3 Institutional Background

As the purpose is to study the association of labour force participation rate on GDP per capita growth in Canada, all datasets are based on Canadian data. All data sets were quarterly and taken from Q1 1995 to Q4 2024.

For investment in inventories, the data had a release date of February 27th, 2026. The status for the dataset is currently set to active. The data was seasonally adjusted at annual rates. Prices are chained (2017) dollars. We only took the entry for investment in inventories. The source of the data was from the National Gross Domestic Product by Income and by Expenditure Accounts (Statistics Canada (2026b)). The National Economic Accounts record the value of GDP from two places: income from production and final expenditure on goods and services produced. The National Gross Domestic Product by Income and Expenditure Accounts are compiled using a variety of data sources and published 60 days following the reference period. They include the economic activities of all establishments, enterprises, households, non-profit institutions serving households, governments, financial corporations, and non-financial corporations operating within the economic territory of Canada. The data are extracted from administrative files and taken from other Statistics Canada surveys and sources. The surveys used to compile the National Gross Domestic Product by Income and by Expenditure Accounts are: the Quarterly Survey of Financial Statements, the monthly Survey of Employment, Payrolls and Hours, the Quarterly Retail Commodity Survey, annual business surveys, banking information, Government Finance Statistics, the balance of payments statistics, and the T4 remuneration file from the Canada Revenue Agency. The National Gross Domestic Product by Income and Expenditure Accounts are estimated by taking high quality data sources available to make benchmark estimates. They take annual business surveys, administrative data files from the CRA and annual household survey files to make the estimates. The data are analyzed for time series consistency and links to economic events as well as coherency.

For labour productivity, the data was taken from "Indexes of business sector labour productivity, unit labour cost and related measures, seasonally adjusted". We took index 2017=100 and specifically the labour productivity (business sector) entry. It was Table 36-10-0206-01. It was seasonally adjusted. The release date was March 4th, 2026. The Canadian Productivity Accounts (CPA) gives data on labour productivity and output, employment, hours worked, labour compensation and unit labour cost (Statistics Canada (2026a)). Quarterly labour productivity series was measured as real gross domestic product per hour worked. The quarterly program of the CPA has three features: estimates are only the measure of labour productivity and related variables, estimates apply to the entire economy and its aggregate business sector, and quarterly estimates

are only available in the form of indexes or growth rates. The CPA is comprised of three programs. The quarterly program provides current estimates of labour productivity and labour costs at the aggregate level. The annual multifactor productivity program provides yearly estimates on multifactor productivity and related measures for the major sectors of the economy and to industries at the national and provincial level. The annual provincial program gives estimates of employment, hours worked, labour productivity, and labour costs at the industry level for each province and territory. All data was extracted from the administrative files and from other surveys: the Labour Force Survey, the Survey of Employment, Payroll and Hours and the Labour Statistics program of the Public Sector Statistics Division. The monthly compensation of employees program of the National Economic Accounts Division gives the benchmarks for the compensation of employees data for the whole economy. In the Canadian System of National Accounts, it is not possible to split the income of self-employed workers between labour and capital, so the compensation of self-employed workers is assumed to be at least equal to that of an employee in the same industry. Labour productivity, hourly compensation, and unit labour cost are indexes derived from chain-weighted output, hours worked, and labour compensation. Three criteria were used to evaluate the quality of the estimates: the quality of the data sources (plus deflators), natures of breaks in the series, and revisions.

For real GDP per capita (chained 2017 dollars), we used Table 36-10-0706-01 from "Gross Domestic Product per capita and other per capita macroeconomic indicators". The release date is February 27, 2026. Similar to investment in inventories, the data was based on National Gross Domestic Product by Income and by Expenditure Accounts. It was also based on Quarterly Demographic Estimates (QDE) (Statistics Canada (2026c)). This program provides quarterly estimates of total population for Canada, provinces and territories. The population for the program includes: Canadian citizens, immigrants with usual place of residence in Canada, immigrants who are abroad, military-based, immigrants at sea, persons with a place of residence in Canada who are asylum claimants, protected persons, members of related groups, or who hold a work, study, or temporary resident permit, government representatives of another country, and residents of another country visiting Canada temporarily. Data are extracted from administrative files and from other Statistics Canada surveys. Population estimates are found taking the population figures from recent census, and adding or subtracting the number of births, deaths, and components of migration. Data are

revised once a year. The data can be divided into two components: births, deaths, and immigration where the data is nearly perfect, and emigration and NPR for which methods have more error.

For labour force participation rate, we took data from the Organization for Economic Co-operation and Development (OECD). They act as a forum for data and analysis where governments from over 100 countries can collaborate to improve economic well-being. They are a United Nations observer. The OECD gets data directly from their member countries' official statistics offices. The OECD then checks for consistency and compares the data to other countries. The data is shown by percentage. It is seasonally adjusted and given quarterly. For labour force participation rate, it was last updated February 16, 2026.

4 Data

In order to analyze the association of labour force participation rate on real GDP per capita, we implement an empirical, quarterly, time-series model from the first quarter of 1995 (1995q1), to the fourth quarter of 2024 (2024q4). As such, we have 30 years of quarterly data, meaning we have 120 observations. Three explanatory variables were chosen: labour force participation rate, investment in inventories, and labour productivity. The response variable is real GDP per capita. The variables' description and summaries are given below.

4.1 Description

- real GDP per capita (Chained 2017 dollars) = rgdppc: Inflation-adjusted GDP per person, expressed in 2017 prices using chain weighting to reflect real living standards. This data is obtained from Statistics Canada (2024c).
- Labour force participation rate = lfpr: The participation rate is the number of labour force participants expressed as a percentage of the population 15 years of age and over. This data is obtained from the Federal Reserve of Economic Data (FRED). Organization for Economic Co-operation and Development (2024).
- Investment in inventories (chained 2017 dollars) (x 1,000,000) = inv: The inflation-adjusted, real change in the quantity of inventories during the period, expressed in 2017 prices using

chain weighting. This data is obtained from Statistics Canada (2024a).

- Labour productivity (business sector) (Index 2017 = 100) = prd: Labour productivity is a measure of real gross domestic product (GDP) per hour worked. This data is obtained from Statistics Canada (2024b).

The final dataset is a CSV file that includes all of the above mentioned variables as columns.

Note: The time variable is not included in the final CSV dataset which was made by merging the data from the variables above. This was done on purpose as it is more convenient to declare the dataset as a quarterly time series dataset in STATA directly.

4.2 Summary Statistics

Variable	Mean	Std. Dev	Min	Max
rgdppc	54204.53	5019.579	40350	61020
lfpr	66.16722	0.971152	62	67.7
inv	9143.608	12486.45	-29778	55297
prd	93.13624	8.68599	75.408	115.319

Table 1: Summary Statistics Table

5 Empirical Framework

5.1 Estimation Strategy

The Empirical Framework features 5 steps

Step 1: Import the data, and declare it to be a quarterly time series. The variable names are as follows:

- rgdppc = real GDP per capita
- lfpr = labour force participation rate
- inv = investment in inventories
- prd =labour productivity

Step 2: Plots and visual evidence on nonstationarity: Plot real GDP per capita, labour force participation rate, investment in inventories, and labour productivity in levels from 1995q1 to 2024q4. This is just a visual check to see if the plots show a trend or a random walk type behaviour.

Step 3: Unit root tests:

We proceed by checking if the variables are $I(0)$ or $I(1)$, that is, we check if the variables are stationary in levels or first differences.

For real GDP per capita, labour force participation rate, investment in inventories, and labour productivity:

- We run an Augmented Dickey-Fuller (ADF) test in levels (include a time trend).
- We run an ADF test in first differences (no trend).
- Based on the results, we classify each series as $I(0)$ or $I(1)$.

Note: We use 4 lags as the data is quarterly.

Step 4: Cointegration check (only if all variables are $I(1)$).

If the results in Step 3 suggest that all variables are $I(1)$, we check whether they are cointegrated using the Engle-Granger two-step approach:

- We regress Y_t on X_t in levels,

$$rgdppc = \beta_0 + \beta_1 lfp_r + \beta_2 inv + \beta_3 prd + ut; \quad (1)$$

and save the residual series $\hat{ut}$. We do not use Newey-West standard errors in this step.

- We Test whether $\hat{ut}$ is stationary using an ADF test (no constant and no trend).

Step 5: Regression of real GDP per capita on labour force participation rate, investment in inventories, and labour productivity, with appropriate transformation (Newey-West).

We run one regression of real GDP per capita on labour force participation rate, investment in inventories, and labour productivity based on our conclusions above:

- If all of real GDP per capita, labour force participation rate, investment in inventories, and

labour productivity are $I(0)$, we estimate the regression in levels:

$$rgdppc = \beta_0 + \beta_1 lfpr + \beta_2 inv + \beta_3 prd + ut \quad (2)$$

- If all of real GDP per capita, labour force participation rate, investment in inventories, and labour productivity are $I(1)$, and we find evidence of cointegration in Step 4, we estimate the regression in levels:

$$rgdppc = \beta_0 + \beta_1 lfpr + \beta_2 inv + \beta_3 prd + ut \quad (3)$$

- If all of real GDP per capita, labour force participation rate, investment in inventories, and labour productivity are $I(1)$ and we do not find evidence of cointegration in Step 4, we estimate the regression in first differences:

$$\Delta rgdppc = \beta_0 + \beta_1 \Delta lfpr + \beta_2 \Delta inv + \beta_3 \Delta prd + ut \quad (4)$$

- If real GDP per capita is $I(1)$ and labour force participation rate, investment in inventories, and labour productivity are $I(0)$, we estimate the regression with real GDP per capita differenced and labour force participation rate, investment in inventories, and labour productivity in levels:

$$\Delta rgdppc = \beta_0 + \beta_1 lfpr + \beta_2 inv + \beta_3 prd + ut \quad (5)$$

- If some variables are $I(1)$ and some variables are $I(0)$, then we estimate the regression with the $I(0)$ variables in levels and the $I(1)$ variables in first differences.

For the regression we estimate in this step, we will report Newey-West HAC standard errors.

6 Results

6.1 Visual Checks

We see in figure A.1 that RGDP Per Capita in levels has an upward trend except for around 2020.

We see in figure A.2 that the labour force participation rate goes up from 1995 to 2003 and then stays steady between 66% to 68% until 2014, with a gradual decrease from 2008. The rate drops

significantly in 2020, and then rises immediately, but never goes above 66%. There is no obvious long run trend.

We can see in figure A.3 that investment in inventories in levels follows a random walk pattern.

We can see in figure A.4 that labour productivity follows a long run upward trend, and a sharp increase and decrease around 2020.

The overall reason for the change in trends during 2020 is due to COVID-19.

6.2 Unit root tests

We see that the results of the Augmented Dickey-Fuller (ADF) test in levels, including a time trend, tells us that only investment in inventories has a significant p-value, and therefore it is $I(0)$, as can be seen in table B.1

Then, we run the Augmented Dickey-Fuller (ADF) test for first differences without a time trend, and notice that real GDP per capita, labour force participation rate, and labour productivity have significant p-values, and therefore they are all $I(1)$, as can be seen in table B.2

Overall:

- Real GDP per capita is $I(1)$
- Labour force participation rate is $I(1)$
- Investment in inventories is $I(0)$
- Labour productivity is $I(1)$

This is consistent with the graphs in A.5 for real GDP per capita, A.6 for labour force participation rate, and A.8 for labour productivity.

6.3 Regression Equation Model

Based on the econometric framework, we do not need to test for cointegration as not all variables are $I(1)$. Also, based on the empirical framework, the final regression equation is modelled as follows:

$$\Delta rgdppc = \beta_0 + \beta_1 \Delta lfpr + \beta_2 inv + \beta_3 \Delta prd + ut \quad (6)$$

After running the regression and getting the results, as can be seen in table B.3, the regression equation is

$$\Delta \hat{r}gdppc = 54.268 + 1806.876\Delta lfpr + 0.007inv + 66.801\Delta prd \quad (7)$$

We see that the fitted values obtained from equation 7 follow closely to the actual values of the first differences in real GDP per capita, as can be seen in figure A.10

Also, based on the regression equation above (7), and the table B.3, we can state the following:

- If there is no change in the labour force participation rate or productivity growth, and inventory investment is zero, the predicted quarterly growth in real GDP per capita is \$54.27.
- For every 1 percentage point increase in the growth of the labour force participation rate, the quarterly growth of real GDP per capita is associated with a \$1,806.88 increase. This is statistically significant at the 5% level (it is in fact even statistically significant at the 1% level).
- For every \$1 million increase in the level of inventory investment, this is associated with the growth of real GDP per capita increasing by \$0.007. This minimal correlation is expected given the variable is measured in levels against a per-capita response. This is statistically significant at the 5% level.
- A 1-unit increase in the growth of the labour productivity Index is associated with a \$66.80 increase in quarterly GDP growth, though this result is not statistically significant at standard levels.

7 Conclusion

This paper investigated the key factors of Canada's real GDP per capita growth from 1995 to 2024 via the labour force sector, specifically examining the association of labour force participation, inventory investment, and labour productivity. The results highlight a strong and positive correlation between the labour force participation rate and economic growth. Our model suggests that a one percentage point increase in participation is associated with a substantial \$1,806.88 increase in quarterly GDP per capita growth, and that it is statistically significant. This suggests that, for

Canada, the labour force participation rate is a major part of the growth of the country's real GDP per capita.

References

- Baldwin, J. R., W. M. Brown, J.-P. Maynard, and D. Zietsma (2004). Catching up and falling behind: the performance of provincial gdp per capita from 1990 to 2003. *Economic Analysis (EA) Research Paper* (024).
- Blinder, A. S. and L. J. Maccini (1991). Taking stock: a critical assessment of recent research on inventories. *Journal of Economic perspectives* 5(1), 73–96.
- Cao, S. and S. Kozicki (2017). Real gdi, productivity, and the terms of trade in canada. *Review of Income and Wealth* 63, S134–S148.
- Hornstein, A. (1998). Inventory investment and the business cycle. *FRB Richmond Economic Quarterly* 84(2), 49–71.
- Maynard, J.-P. (2007, Aug). A comparison of gdp per capita in canada and the united states from 1994 to 2005. Insights on the Canadian Economy 2007016e, Statistics Canada, Economic Analysis Division.
- Organization for Economic Co-operation and Development (2024). Infra-annual labor statistics: Labor force participation rate total: 15 years or over for canada [lracttttcaq156s]. Retrieved from FRED, Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/series/LRACTTTTCAQ156S>. Labour Force Participation Rate. Accessed: 2026-04-06.
- Smith, P. (2024). Accounting for the decline of canada’s real gdp per capita since mid-2022. *International Productivity Monitor* (46), 83–100.
- Statistics Canada (2024a). Table 36-10-0104-01: Gross domestic product, expenditure-based, canada, quarterly. <https://doi.org/10.25318/3610010401-eng>. Investment in Inventories. [Data table]. Accessed: 2026-04-06.
- Statistics Canada (2024b). Table 36-10-0206-01: Indexes of business sector labour productivity, unit labour cost and related measures, seasonally adjusted. <https://doi.org/10.25318/3610020601-eng>. Labour Productivity (business sector). [Data table]. Accessed: 2026-04-06.

Statistics Canada (2024c). Table 36-10-0706-01: Gross domestic product per capita and other per capita macroeconomic indicators. <https://doi.org/10.25318/3610070601-eng>. Real GDP Per Capita (chained 2017 dollars). [Data table]. Accessed: 2026-04-06.

Statistics Canada (2026a). Labour productivity measures - national (quarterly) (lpm). <https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=5042>. Accessed: 2026-04-06.

Statistics Canada (2026b). National gross domestic product (gdp) by income and by expenditure accounts. <https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=1901>. Accessed: 2026-04-06.

Statistics Canada (2026c). Quarterly demographic estimates (qde). Accessed April 6, 2026.

Wölfl, A. and D. Hajkova (2007). Measuring multifactor productivity growth.

A Appendix A: Graphs of Key Variables

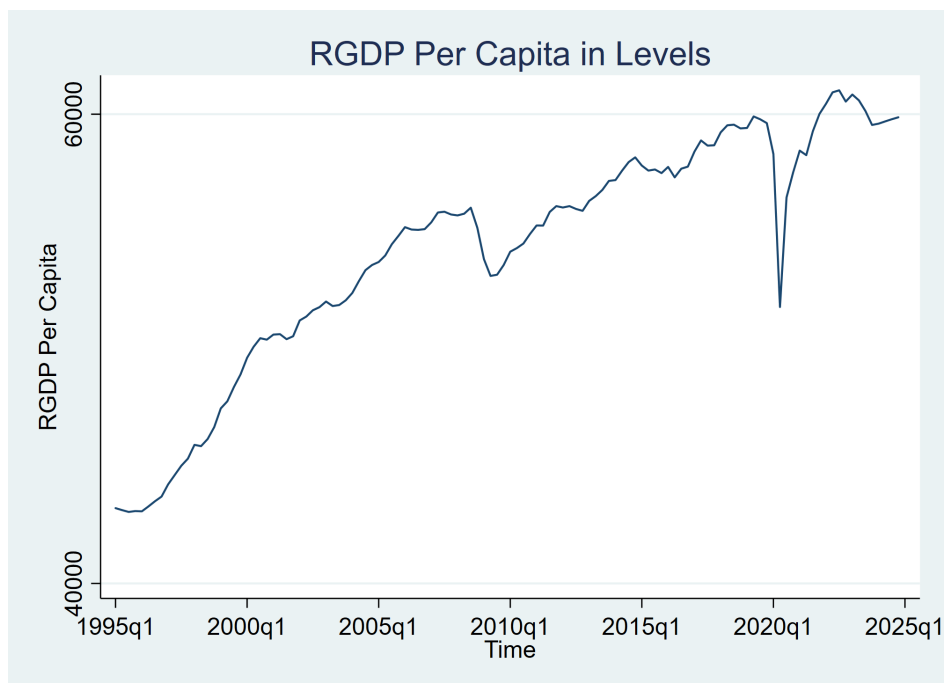


Figure A.1: Real GDP per capita in levels

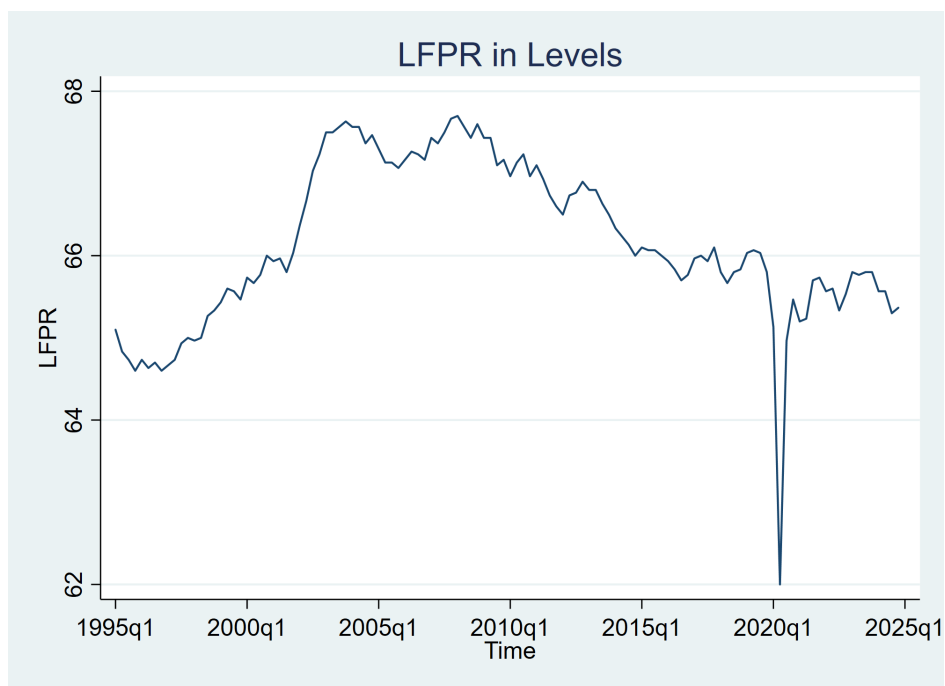


Figure A.2: Labour force participation rate in levels

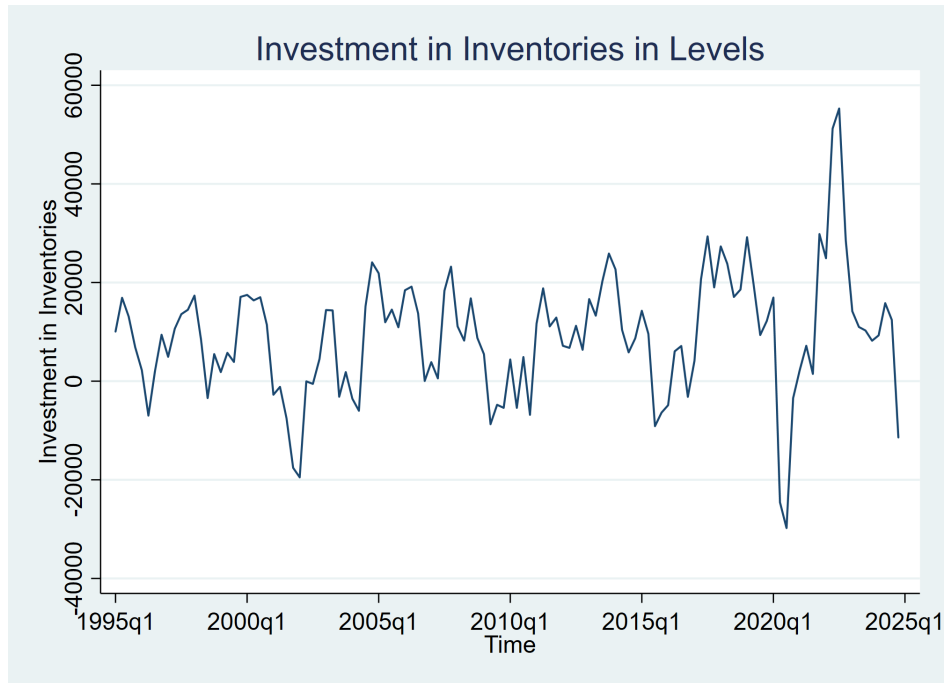


Figure A.3: Investment in inventories in levels

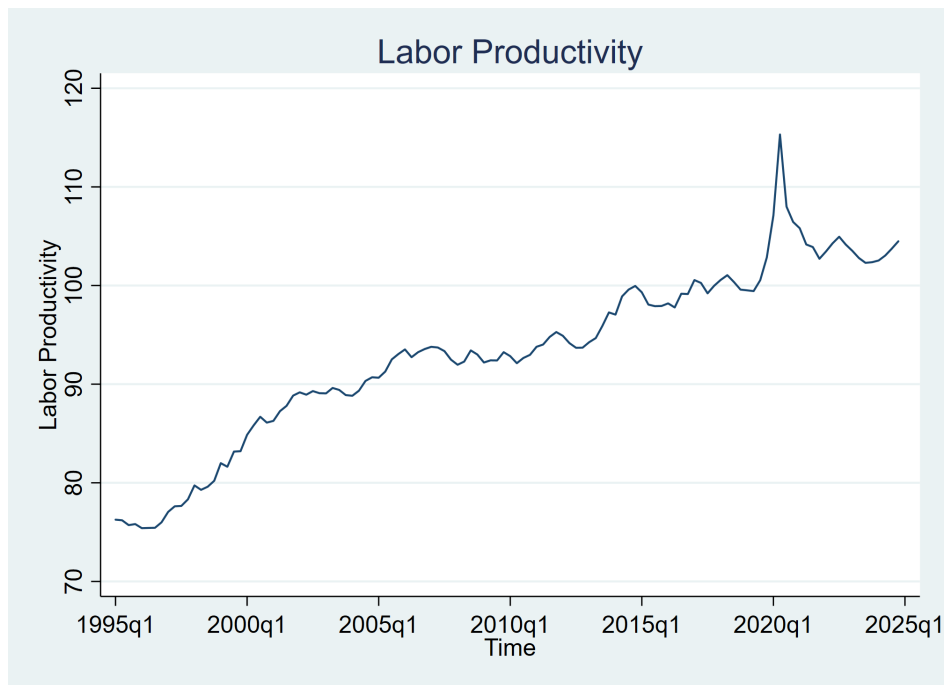


Figure A.4: Labour productivity in levels

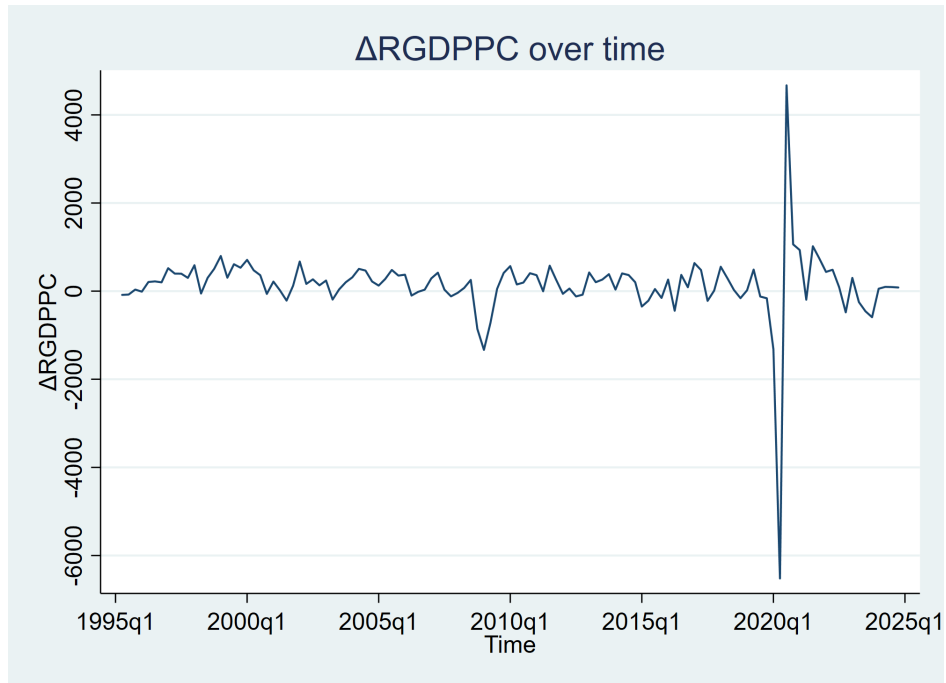


Figure A.5: Δ Real GDP per capita

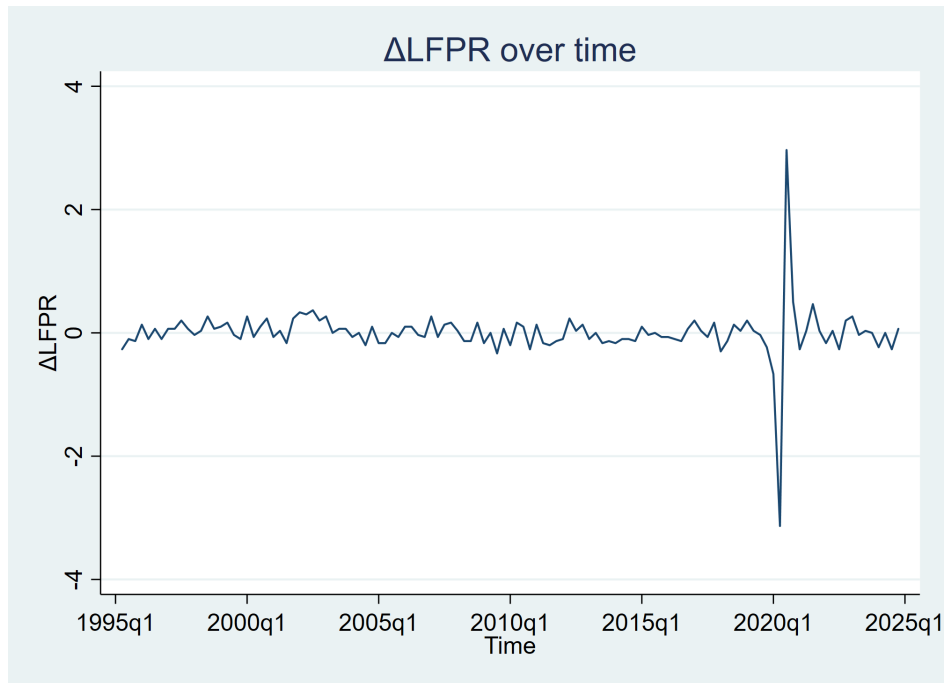


Figure A.6: Δ Labour force participation rate

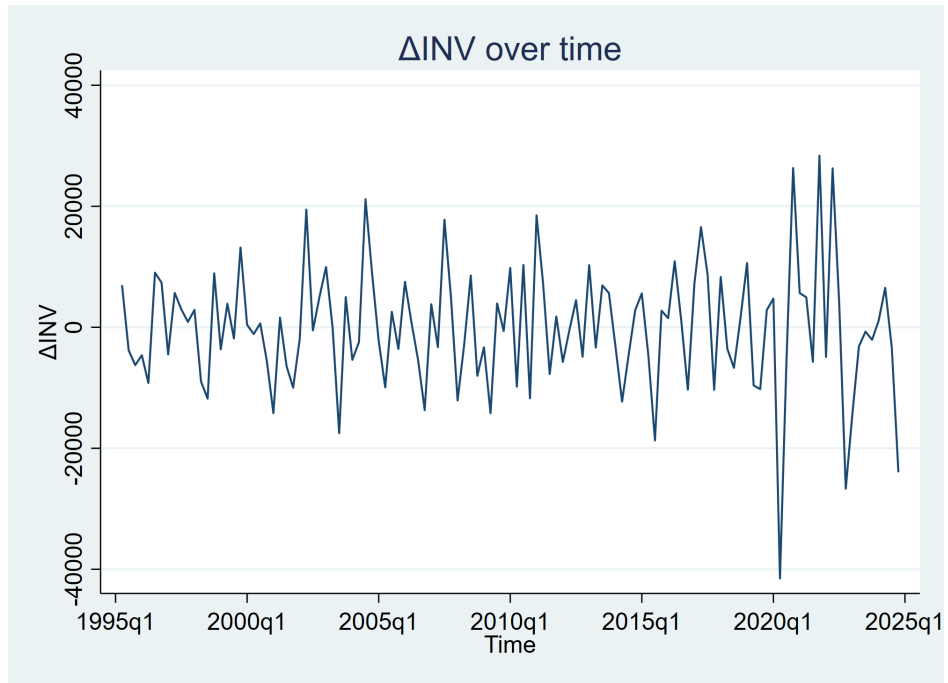


Figure A.7: Δ Investment in inventories

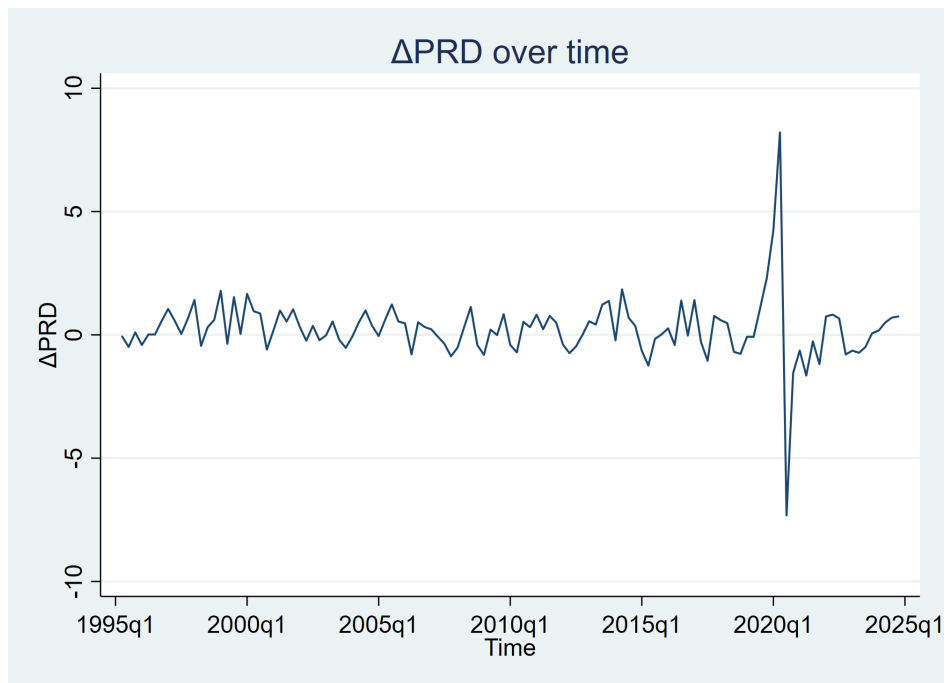


Figure A.8: Δ Labour productivity

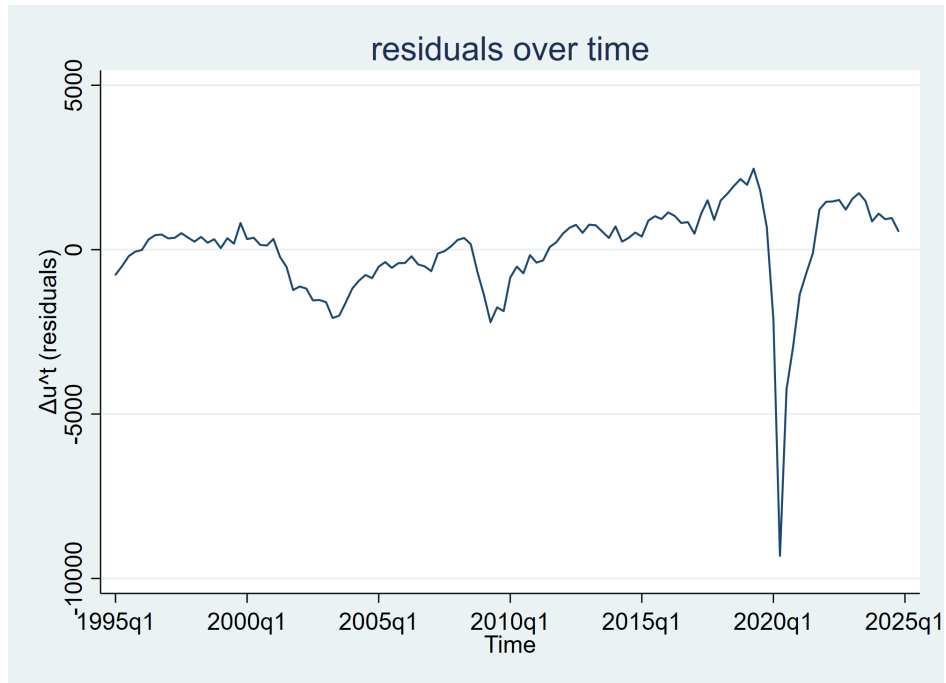


Figure A.9: Residuals over time

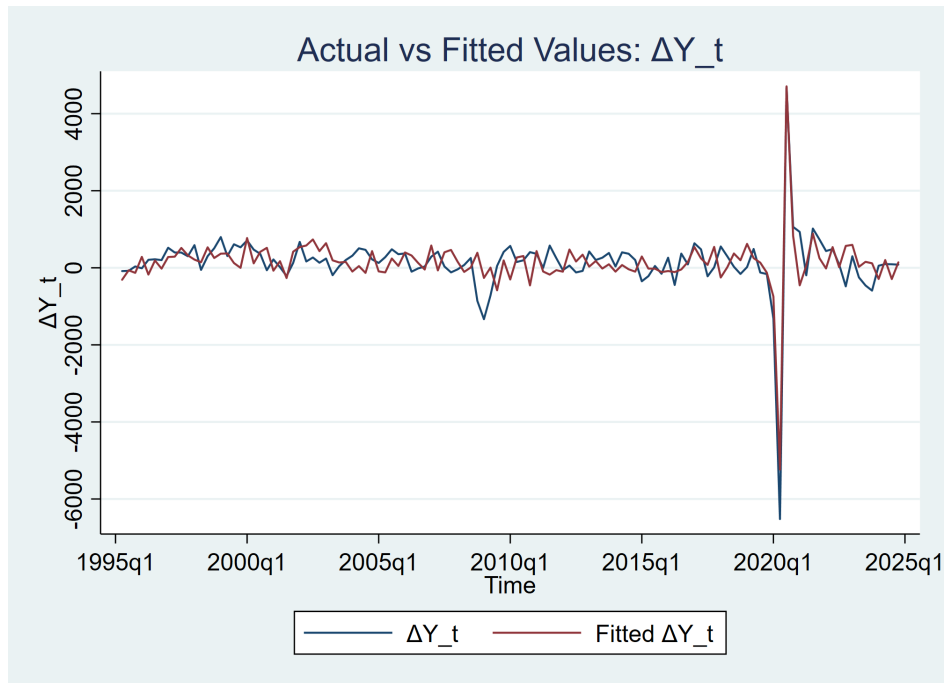


Figure A.10: Actual vs fitted values for Δ real GDP per capita

B Appendix B: Test Result Tables

	Test Statistic	5% Critical Value	p-value	Is it I(0)?
rgdppc	-2.4953	-3.488	0.3367	NO
lfpr	-1.892	-3.488	0.6588	NO
inv	-5.124	-3.488	0.0001	YES
prd	-2.690	-3.488	0.2401	NO

Table B.1: LEVELS - ADF TEST RESULTS

	Test Statistic	5% Critical Value	p-value	Is it I(1)?
rgdppc	-5.070	-2.889	0.0000	YES
lfpr	-5.974	-2.889	0.0000	YES
inv	-6.135	-2.889	0.0000	N/A
prd	-5.731	-2.889	0.0000	YES

Table B.2: FIRST DIFFERENCES - ADF TEST RESULTS

Table B.3: Regression Results

	(1) 1st diff in Real GDP Per Capita
1st diff in Labor Force Participation Rate	1806.876*** (185.276)
Investment in Inventories (x1,000,000) (Chained 2017 dollars)	0.007** (0.003)
1st diff in Labor Productivity	66.801 (64.773)
Constant	54.268 (61.138)
Observations	119

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$